WiFi Training Program

Module 3 - Assignment

1. What are the different 802.11 PHY layer standards? Compare their characteristics?

The IEEE 802.11 family defines multiple PHY layer standards, each introducing improvements in frequency band, modulation technique, data rate, and spatial stream support. The following are the principal standards:

802.11 (1997): The original standard operating in the 2.4 GHz band using either DSSS or FHSS modulation. Maximum data rate of 2 Mbps. Rarely encountered today.

802.11b (1999): Operates at 2.4 GHz using DSSS with CCK (Complementary Code Keying). Maximum data rate of 11 Mbps. Widely deployed in early Wi-Fi but now obsolete. Single spatial stream, 20 MHz channel width, range up to 100 m indoors.

802.11a (1999): Operates at 5 GHz using OFDM. Maximum data rate of 54 Mbps across 8 non-overlapping 20 MHz channels. Shorter range than 802.11b due to higher frequency but less congestion. Single spatial stream.

802.11g (2003): Operates at 2.4 GHz using OFDM (same as 802.11a) with backward compatibility with 802.11b. Maximum data rate of 54 Mbps. Single spatial stream, 20 MHz channels.

802.11n (Wi-Fi 4, 2009): Operates in both 2.4 GHz and 5 GHz. Introduces MIMO (up to 4 spatial streams), 40 MHz channel bonding, and higher-order QAM. Maximum data rate of 600 Mbps. Adds Short Guard Interval (400 ns) and frame aggregation (A-MPDU, A-MSDU).

802.11ac (Wi-Fi 5, 2013): Operates exclusively in 5 GHz. Introduces MU-MIMO (downlink, up to 4 users), 80 MHz and 160 MHz channels, up to 8 spatial streams, and 256-QAM. Maximum theoretical data rate of 6.9 Gbps (Wave 2).

802.11ax (Wi-Fi 6/6E, 2019/2021): Operates in 2.4 GHz, 5 GHz, and (6E) 6 GHz. Introduces OFDMA, 1024-QAM, BSS Coloring, TWT, uplink MU-MIMO (up to 8 users), and 8 spatial streams. Maximum theoretical rate of 9.6 Gbps.

802.11be (Wi-Fi 7, 2024): Operates across 2.4 GHz, 5 GHz, and 6 GHz simultaneously via Multi-Link Operation (MLO). Introduces 320 MHz channels, 4096-QAM, 16 spatial streams, Multi-RU assignment, and preamble puncturing. Maximum theoretical rate of 46 Gbps.

In summary, each generation improves over the previous by increasing channel width, modulation density, spatial stream count, and multi-user capabilities, while also moving to less congested frequency bands.

2. What are DSSS and FHSS? How do they work?

DSSS (Direct Sequence Spread Spectrum): DSSS spreads the data signal across a wider frequency band by multiplying it with a high-rate chipping code (called a Barker code or similar pseudo-random sequence). Each data bit is represented by multiple chips (e.g., 11 chips per bit in 802.11b). The receiver uses the same chipping code to de-spread the signal and recover the original data. Spreading the signal makes it more resistant to narrowband interference. If interference affects part of the spectrum, the rest still carries the data. Used in 802.11 (legacy) and 802.11b. Advantage: More robust to interference and multipath fading. Disadvantage: Less spectrally efficient than OFDM.

FHSS (Frequency Hopping Spread Spectrum): FHSS transmits data by rapidly hopping the carrier frequency across multiple channels in a pseudorandom sequence known to both transmitter and receiver. The transmitter and receiver synchronize their hopping patterns. If a particular frequency is jammed or has interference, the next hop moves to a clean frequency. Used in the original 802.11 (1997) standard. Advantage: Highly resistant to narrowband jamming and interception. Disadvantage: Lower throughput compared to DSSS; not used in modern Wi-Fi.

Key Difference: DSSS spreads across all frequencies simultaneously using coding; FHSS hops between frequencies over time.

3. How do modulation schemes work in the PHY layer? Compare different modulation schemes and their performance across various Wi-Fi standards.

Modulation is the process by which digital data bits are encoded onto a radio carrier wave by varying one or more of the wave's properties: amplitude, phase, or frequency. The PHY layer is responsible for selecting and applying the appropriate modulation scheme based on current channel conditions.

BPSK (Binary Phase Shift Keying): The simplest modulation scheme, encoding 1 bit per symbol by shifting the carrier phase between two states (0 and 180 degrees). Highly robust with very low SNR requirements. Used for the lowest MCS (MCS 0) in all Wi-Fi standards and for management and control frames that must be decodable at maximum range.

QPSK (Quadrature Phase Shift Keying): Encodes 2 bits per symbol using four phase states (0, 90, 180, 270 degrees). Doubles the data rate of BPSK with a modest increase in required SNR. Used in MCS 1 and MCS 2 across 802.11a/g/n/ac/ax.

16-QAM (Quadrature Amplitude Modulation): Combines amplitude and phase modulation to encode 4 bits per symbol across 16 constellation points. Requires a better SNR than QPSK but achieves twice the data rate. Used in MCS 3 and MCS 4 in 802.11n/ac/ax.

64-QAM: Encodes 6 bits per symbol across 64 constellation points. The highest order modulation in 802.11a/g. Used in MCS 5, 6, and 7 in 802.11n, providing up to 65 Mbps per spatial stream in a 20 MHz channel. Requires good SNR (approximately 25 dB or higher).

256-QAM: Encodes 8 bits per symbol across 256 constellation points. Introduced in 802.11ac (Wi-Fi 5). Provides a 33 percent throughput gain over 64-QAM for the same channel conditions. Requires approximately 30 dB SNR. Used in MCS 8 and MCS 9.

1024-QAM: Encodes 10 bits per symbol. Introduced in 802.11ax (Wi-Fi 6). Delivers a 25 percent throughput gain over 256-QAM. Requires high SNR (approximately 35 dB) and is practical only at short to medium range. Used in MCS 10 and MCS 11.

4096-QAM: Encodes 12 bits per symbol. Introduced in 802.11be (Wi-Fi 7). Provides a 20 percent throughput gain over 1024-QAM. Requires excellent SNR (approximately 40 dB or higher) and is practical only at very close range in clean RF environments.

The PHY layer dynamically selects the modulation scheme through a process called Link Adaptation or Rate Control. The transmitter continuously monitors channel quality (SNR, PER) and selects the highest-order modulation that the current conditions can reliably support. As the client moves farther from the AP or encounters more interference, the MCS index drops to a more robust lower-order modulation. This adaptive process is why Wi-Fi connection speeds appear to vary dynamically.

4. What is the significance of OFDM in WLAN? How does it improve performance?

OFDM stands for Orthogonal Frequency Division Multiplexing. It is the core modulation technique used in 802.11a/g/n/ac/ax and is the reason modern Wi-Fi can achieve high data rates reliably.

OFDM divides a wide channel into many narrow, orthogonal subcarriers (e.g., 52 subcarriers in a 20 MHz channel). Each subcarrier is modulated independently with BPSK, QPSK, or QAM. The subcarriers are orthogonal to each other, meaning they do not interfere even though they overlap in frequency.

Significance and Performance Improvements: Multipath Resistance: In indoor environments, signals reflect off walls and arrive at the receiver at different times (multipath). OFDM's narrow subcarriers have a long symbol duration, making them far less sensitive to multipath interference than single-carrier systems like DSSS. High Spectral Efficiency: Multiple subcarriers carry data simultaneously, making full use of the available bandwidth. Scalability: OFDM supports variable modulation per subcarrier, allowing adaptation to channel conditions. Guard Interval: A cyclic prefix (guard interval) is added to each symbol to absorb multipath echoes without causing inter-symbol interference (ISI). Foundation for OFDMA: OFDM was extended to OFDMA in 802.11ax, allowing multiple users to share subcarriers simultaneously.

Impact: OFDM enabled 802.11a/g to reach 54 Mbps vs 11 Mbps for 802.11b (DSSS), and it remains the foundation of all modern Wi-Fi standards.

5. How are frequency bands divided for Wi-Fi? Explain different bands and their channels.

Wi-Fi operates in three main frequency bands: 2.4 GHz, 5 GHz, and 6 GHz (Wi-Fi 6E and Wi-Fi 7). Each band has different characteristics and is divided into channels.

2.4 GHz Band: Range 2.400 to 2.4835 GHz. Total Channels: 14 (varies by country; US uses 1 to 11). Non-overlapping Channels: Only 3 (Channels 1, 6, 11) with 20 MHz width. Characteristics: Longer range, better wall penetration, but more congested (used by microwaves, Bluetooth, baby monitors). Standards: 802.11b/g/n/ax.

5 GHz Band: Range 5.150 to 5.850 GHz. Total Channels: Up to 25 non-overlapping 20 MHz channels (varies by region and DFS rules). Channel Widths: 20, 40, 80, 160 MHz. Characteristics: Shorter range, less congestion, higher throughput. Standards: 802.11a/n/ac/ax. DFS Channels: Some channels (52 to 140) require Dynamic Frequency Selection to avoid radar interference.

6 GHz Band (Wi-Fi 6E / Wi-Fi 7): Range 5.925 to 7.125 GHz. Total Channels: 59 non-overlapping 20 MHz channels (up to 7 x 160 MHz channels). Characteristics: Least congested, highest throughput, shortest range. Standards: 802.11ax (Wi-Fi 6E), 802.11be (Wi-Fi 7). Only newer devices support this band.

Key Principle: Higher frequency equals shorter range but more available spectrum and less interference. Lower frequency equals longer range but more crowded spectrum.

6. What is the role of Guard Intervals in WLAN transmission? How does a short Guard Interval improve efficiency?

A Guard Interval (GI) is a period of silence deliberately inserted between successive OFDM symbols during transmission. Its purpose is to prevent Inter-Symbol Interference (ISI), which occurs when a reflected copy of a previous symbol (caused by multipath propagation) arrives at the receiver at the same time as the next symbol, corrupting the data.

How the Guard Interval Works: In a multipath environment, a transmitted signal bounces off walls, ceilings, furniture, and other objects and arrives at the receiver via multiple paths with different propagation delays. If the delay spread of the channel exceeds the symbol boundary, the tail of a reflected previous symbol overlaps with the beginning of the current symbol, causing ISI. The guard interval creates a time buffer at the start of each symbol that is longer than the maximum expected delay spread of the channel, absorbing the echoes before the receiver begins sampling the useful symbol content.

The guard interval is implemented as a Cyclic Prefix (CP): a copy of the last portion of the OFDM symbol is prepended to the symbol itself. The receiver discards the guard interval portion and decodes only the clean symbol body. This ensures that even if multipath echoes persist into the guard interval period, they do not corrupt the data-carrying portion of the symbol.

Standard Guard Interval (800 ns): The default guard interval in 802.11a/g/n/ac is 800 nanoseconds (0.8 microseconds). This is sufficient to handle the delay spreads encountered in most indoor environments, which typically range from 50 to 300 ns. The total OFDM symbol duration is 3.2 microseconds (data) plus 0.8 microseconds (GI) equals 4.0 microseconds per symbol.

Short Guard Interval (400 ns): 802.11n introduced the optional Short Guard Interval (SGI) of 400 nanoseconds, half the standard GI duration. By reducing the guard interval from 800 ns to 400 ns, the symbol period is shortened from 4.0 microseconds to 3.6 microseconds. This reduction means more symbols can be transmitted per second, directly increasing the data throughput by approximately 11 percent (from 4.0 to 3.6 microseconds, a ratio of 4.0/3.6 = 1.111).

For example, 802.11n with MCS 7 (64-QAM, 5/6 coding rate, 1 stream, 20 MHz) achieves 65 Mbps with standard GI and 72.2 Mbps with short GI. The SGI is only safe to use when the channel delay spread is well below 400 ns, meaning it is appropriate for short-range, line-of-sight links in low-multipath environments. In high-multipath environments such as large warehouses or buildings with metal structures, using SGI can cause ISI and actually reduce throughput due to increased retransmissions.

802.11ax (Wi-Fi 6) further extends this concept by introducing multiple GI options: 0.8 microseconds (standard), 1.6 microseconds, and 3.2 microseconds, along with a much longer symbol duration of 12.8 microseconds. The longer symbol duration of 802.11ax inherently reduces the proportional overhead of the GI, improving efficiency even with the standard 0.8 microsecond GI.

7. Describe the structure of an 802.11 PHY layer frame. What are its key components?

An 802.11 PHY layer frame, formally called a PPDU (Physical Protocol Data Unit), is the complete unit of data transmitted over the wireless medium. It encapsulates the MAC layer frame (MPDU) within PHY-specific headers and preambles that enable the receiver to synchronize, detect the signal, determine the transmission parameters, and decode the data. While the exact structure varies across 802.11 generations, the fundamental components are as follows:

PLCP Preamble: The preamble is the first section of the PPDU and consists of known training sequences transmitted at a fixed, low rate that all compatible devices can detect. It serves several critical functions. The Short Training Field (STF) allows the receiver to detect the presence of a signal, perform automatic gain control (AGC) to set the receive amplifier to an appropriate level, perform coarse frequency offset correction, and achieve timing synchronization. The Long Training Field (LTF) allows the receiver to perform precise channel estimation and fine frequency offset correction, providing the channel response information needed to equalize the received signal and decode the data. In MIMO systems (802.11n and later), multiple LTFs are transmitted (one per spatial stream) to enable per-stream channel estimation.

PLCP Header (Signal Field): The signal field follows the preamble and carries the parameters the receiver needs to correctly decode the data field. It is always transmitted at a known, low, mandatory rate (6 Mbps BPSK in 802.11a/g/n/ac) so that all devices can read it. The signal field contains the MCS index (or data rate), channel width, number of spatial streams, frame length (number of octets in the PSDU), STBC and FEC type (BCC or LDPC), and guard interval duration. In newer standards (802.11n and later), the signal field is split into multiple sections: for example, 802.11n includes L-SIG (legacy, backward-compatible signal) and HT-SIG (High Throughput signal), while 802.11ax includes L-SIG, RL-SIG (repeated L-SIG for detection), HE-SIG-A (common parameters), and HE-SIG-B (per-user RU allocation in MU-PPDU).

PSDU (Physical Service Data Unit) / Data Field: The PSDU is the payload of the PPDU and contains the MAC layer frame(s) (MPDU or A-MPDU). It is transmitted at the negotiated data rate using the MCS specified in the signal field. The data field includes a SERVICE field (16 bits, used for scrambler initialization and signal extension), the MPDU or aggregated MPDUs (A-MPDU), PADDING bits to align the total number of bits to an integer number of OFDM symbols, and TAIL bits used to flush the convolutional encoder (in BCC-coded frames).

FCS (Frame Check Sequence): Although the FCS is technically part of the MAC layer frame (MPDU) rather than a separate PHY component, it is included in the PSDU and protects the integrity of the MAC frame payload. It is a 32-bit CRC calculated over the MAC header and frame body. The PHY layer does not add its own separate FCS; integrity at the PPDU level is ensured by the training fields enabling accurate channel equalization and FEC protecting the data bits.

In summary, the preamble enables signal detection and channel estimation, the signal field conveys decoding parameters, and the PSDU carries the actual MAC data, all working together to deliver reliable, high-rate wireless transmission.

8. What is the difference between OFDM and OFDMA?

OFDM (Orthogonal Frequency Division Multiplexing) and OFDMA (Orthogonal Frequency Division Multiple Access) share the same fundamental physical layer technology of dividing a channel into many narrow orthogonal subcarriers. The critical difference between them is how those subcarriers are allocated across users.

OFDM - Single User at a Time: In OFDM as used in 802.11a/g/n/ac, the entire set of available subcarriers is assigned to a single user for the duration of each TXOP (Transmission Opportunity). When the AP transmits to one client, all subcarriers carry that client's data. When the AP then transmits to a different client, all subcarriers carry that client's data. Users are served sequentially in time, taking turns accessing the entire channel. This works well when each client has enough data to fill the entire channel, but is wasteful when a client has only a small amount of data (such as a VoIP packet or a sensor reading), because it still monopolizes the entire channel bandwidth for the duration of the transmission.

OFDMA - Multiple Users Simultaneously: In OFDMA as used in 802.11ax (Wi-Fi 6) and later, the available subcarriers are grouped into Resource Units (RUs) of varying sizes (26, 52, 106, 242, 484, or 996 tones in a single 20 MHz channel or wider). Different RUs can be assigned to different clients within a single TXOP, allowing the AP to serve multiple clients simultaneously within one PPDU transmission. A client that only needs a small amount of data is assigned a small RU (e.g., 26 tones), while a client with heavy traffic may be assigned a larger RU (e.g., 242 tones or more). Multiple clients share the same time slot, each using their assigned portion of the frequency spectrum.

OFDMA works for both downlink (DL-OFDMA, AP to clients) and uplink (UL-OFDMA, clients to AP via Trigger frame). The AP coordinates UL-OFDMA by sending a Trigger frame that specifies which clients should transmit, on which RUs, and with what power level. All triggered clients then transmit simultaneously.

Key Impact: OFDM is single-user per TXOP. OFDMA is multi-user per TXOP. This distinction makes OFDMA dramatically more efficient in networks with many low-bandwidth devices, reducing the overhead of channel access contention per client, lowering average latency, and increasing overall network throughput per unit time.

9. What is the difference between MIMO and MU-MIMO?

MIMO (Multiple Input Multiple Output) and MU-MIMO (Multi-User Multiple Input Multiple Output) both use multiple antennas at the transmitter and receiver to improve wireless performance, but they differ fundamentally in whether the multiple spatial streams serve one device or multiple devices simultaneously.

MIMO - Single User, Multiple Streams: In MIMO (also called SU-MIMO, Single-User MIMO), multiple antennas at the AP and a single client are used to transmit multiple independent data streams to that one client simultaneously over the same frequency channel. This is called Spatial Multiplexing. For example, a 2x2 MIMO system (2 transmit antennas, 2 receive antennas) can transmit 2 independent spatial streams to one client at once, doubling the data rate compared to a single-antenna system. A 4x4 MIMO system can deliver up to 4 streams. MIMO was introduced in 802.11n (up to 4 streams) and extended in 802.11ac (up to 8 streams). The key constraint of MIMO is that all spatial streams go to the same client at the same time. While that client is being served, all other clients must wait.

MU-MIMO - Multiple Users, Multiple Streams: MU-MIMO extends the spatial multiplexing concept so that the AP can transmit to multiple clients simultaneously, each on their own spatial stream(s), using the same frequency channel and the same time slot. The AP uses beamforming (directing the signal toward each client using phase shifting across antennas) to spatially separate the streams intended for different clients, minimizing inter-stream interference. Downlink MU-MIMO (DL MU-MIMO) was introduced in 802.11ac Wave 2 (up to 4 simultaneous users). 802.11ax extended this to 8 simultaneous users in both downlink and uplink (UL MU-MIMO).

For MU-MIMO to work effectively, the AP must obtain Channel State Information (CSI) from each client through a sounding process. The AP sends Null Data Packets (NDPs) and each client responds with a compressed beamforming feedback matrix. The AP uses this per-client channel information to compute the beamforming steering matrix that directs energy toward each intended client while creating nulls in the directions of other clients.

Key Difference Summary: MIMO (SU-MIMO) increases throughput for one client by sending multiple streams to that client. MU-MIMO increases network throughput by serving multiple clients simultaneously in the same TXOP. SU-MIMO benefits high-bandwidth individual clients; MU-MIMO benefits overall network capacity in multi-client environments. Both are used together in modern

Wi-Fi: the AP chooses SU-MIMO when one client needs maximum throughput, and MU-MIMO when it is more efficient to serve several clients at once.

10. What are PPDU, PLCP, and PMD in the PHY layer?

PPDU (Physical Protocol Data Unit): The PPDU is the complete frame format at the PHY layer. It is what actually gets transmitted over the air. It consists of the PLCP Preamble, PLCP Header, and the PSDU (the MAC data). Every Wi-Fi transmission on the medium is a PPDU.

PLCP (Physical Layer Convergence Protocol): The PLCP is the upper sublayer of the PHY layer. It acts as an interface between the MAC layer and the physical medium. Its responsibilities are: Framing: Adds preamble and header to the MAC data (MPDU) to form the PPDU. Rate Negotiation: Signals the data rate, channel width, and MCS to the receiver in the PLCP header. Clear Channel Assessment (CCA): Monitors the medium and tells the MAC layer whether the channel is busy or free (used in CSMA/CA). Provides a standard interface so the same MAC can work with different physical mediums.

PMD (Physical Medium Dependent): The PMD is the lower sublayer of the PHY layer. It handles the actual transmission and reception of bits over the radio medium. Its responsibilities are: Modulation and Demodulation: Converts digital bits into analog radio signals (and back). Encoding: Applies forward error correction (FEC) such as convolutional coding or LDPC. Frequency and Power: Controls the transmit frequency, power level, and antenna operations. The PMD sublayer is what differs between 802.11a (OFDM at 5 GHz), 802.11b (DSSS at 2.4 GHz), etc.

Summary: PPDU is the full transmitted frame. PLCP prepares and frames the data and manages CCA. PMD actually sends and receives the bits over the air using radio signals.

11. What are the types of PPDU? Explain the PPDU frame format across different Wi-Fi generations.

Different 802.11 standards define different PPDU formats, each adding fields to support new capabilities like MIMO, wider channels, and multi-user transmission.

1. Non-HT PPDU (Legacy) - 802.11a/b/g: Format: Preamble (STF + LTF) followed by Signal Field (SIGNAL) followed by Data. Supports up to 54 Mbps (802.11a/g) or 11 Mbps (802.11b). Single spatial stream only, no MIMO.

2. HT PPDU (High Throughput) - 802.11n (Wi-Fi 4): Format: L-STF, L-LTF, L-SIG, HT-SIG, HT-STF, HT-LTF(s), Data. Adds HT-SIG field to signal MIMO parameters (number of spatial streams, channel width, MCS index). Supports up to 600 Mbps with 4 spatial streams and 40 MHz channels. Two sub-types: HT-Mixed (backward compatible) and HT-Greenfield (no legacy support).

3. VHT PPDU (Very High Throughput) - 802.11ac (Wi-Fi 5): Format: L-STF, L-LTF, L-SIG, VHT-SIG-A, VHT-STF, VHT-LTF(s), VHT-SIG-B, Data. VHT-SIG-A carries MCS, number of streams, channel width (up to 160 MHz). VHT-SIG-B carries per-user information for MU-MIMO. Supports up to 3.5 Gbps with 8 spatial streams.

4. HE PPDU (High Efficiency) - 802.11ax (Wi-Fi 6): Format: L-STF, L-LTF, L-SIG, RL-SIG, HE-SIG-A, HE-SIG-B, HE-STF, HE-LTF(s), Data. HE-SIG-B carries per-user Resource Unit (RU) allocation for OFDMA. Four sub-types: HE SU (single user), HE MU (multi-user OFDMA), HE TB (trigger-based uplink), HE ER SU (extended range). Supports up to 9.6 Gbps.

5. EHT PPDU - 802.11be (Wi-Fi 7): Adds Multi-Link Operation (MLO) support. Supports 320 MHz channels and 4096-QAM. Up to 46 Gbps theoretical throughput.

12. How is the data rate calculated?

The data rate in Wi-Fi is not a fixed value; it is dynamically calculated based on several physical layer parameters. The formula for data rate is: Data Rate = (Subcarriers x Bits per Symbol x Coding Rate x Spatial Streams) divided by Symbol Duration.

Where: Number of Data Subcarriers depends on channel width (e.g., 20 MHz OFDM produces 48 data subcarriers; 40 MHz produces 108 subcarriers). Bits per Symbol (Modulation): BPSK=1, QPSK=2, 16-QAM=4, 64-QAM=6, 256-QAM=8, 1024-QAM=10. Coding Rate (FEC Rate): Forward Error Correction adds redundancy. Common rates: 1/2, 2/3, 3/4, 5/6. A rate of 3/4 means 75 percent of bits are data and 25 percent are error correction. Spatial Streams (MIMO): Each additional spatial stream multiplies the data rate. Symbol Duration: Typically 4 microseconds in 802.11a/g/n/ac (3.2 microseconds OFDM plus 0.8

microseconds GI); 12.8 microseconds in 802.11ax (with 0.8, 1.6, or 3.2 microsecond GI options).

Example: 802.11n, 20 MHz, 64-QAM, 3/4 coding, 1 stream, short GI: Data subcarriers: 52. Bits per subcarrier: 6 (64-QAM). Coding rate: 3/4. Spatial streams: 1. Symbol duration: 3.6 microseconds (with short GI). Data rate = (52 x 6 x 3/4 x 1) divided by 3.6 microseconds = 234 divided by 3.6 = 65 Mbps.

MCS Index: In practice, the combination of modulation and coding rate is represented as an MCS (Modulation and Coding Scheme) index (e.g., MCS 0 = BPSK 1/2, MCS 7 = 64-QAM 5/6 in 802.11n). The AP and client negotiate the highest MCS that the current SNR can support, and this determines the actual link speed shown in the Wi-Fi connection properties.